# ECON 8803: Replication Exercise

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May 1, 2026

**Abstract**

Katovich set out to examine the causal effect of different types of energy-generating infrastructure on avian ecosystems across the United States. He specifically studied the impact of shale gas wells and wind turbine electricity production as both saw increases by orders of magnitude from 2000 to 2020. The study uses three primary data sources: Rystad Energy's U-Cube Platform for geolocated shale field data for over 240,000 wells, the U.S. Wind Turbine Database from the US Department of Energy covering 115,000 turbines, and the National Audubon Society's Annual Christmas Bird Count (CBC) longitudinal data to identify circles 24km in diameter to measure bird population and biodiversity.

Katovich defined treated CBC circles as those with a well or turbine built within 5km of the circle boundary. The key outcome is the number of birds counted in circle $c$ in year $t$ measured with the group-time difference-in-differences specification with standard errors clustered at the circle-level.

This analysis found that shale wells reduced bird population counts by 15% even when accounting for location and year fixed effects along with no measurable effect from installing wind turbines. Additionally, shale wells had a heterogeneous impact on birds, with greater negative effects within important bird habitats. The American press has spilled a disproportionate amount of ink on the impact of wind turbines on bird habitats despite much stronger evidence that fracking or shale are meaningfully harmful to birds.

# 1 Pre-Analysis

This analysis is a replication of Erik Katovich's 2023 paper "Quantifying the Effects of Energy Infrastructure on Bird Populations and Biodiversity." The author examines the impact of shale oil and gas production and wind energy generation on bird populations across the United States. From 2000 to 2020, both shale gas production and onshore wind energy capacity both increased by orders of magnitude: 20-fold for shale and 48-fold for wind. The built infrastructure required impacts natural ecosystems, from direct effects around land use to indirect effects like groundwater contamination and various types of pollution.

Previous work of smaller scale found effects of both shale and wind production on bird populations. A 2016 study conducted by Shaffer and Buhl found that 7 of the 9 grassland bird species studied displayed avoidant behavior within 300m of wind turbines. Additionally, a 2023 study from Burg et al of four key species found that the Savannah sparrow specifically showed a negative response to oil and gas development in North Dakota.

Katovich significantly expanded the previous literature by combining bird population estimates from the National Audubon Society's Christmas Bird Count (CBC) with wind turbine data from the United States Geological Survey (USGS) and shale oil and gas well data from Rystad Energy. The CBC evaluates bird counts annually by estimating both the count of birds and the species observed within a 24.1km (15 mile circle). The USGS has comprehensive wind turbine data for all turbines built since 1980, and Rystad has data for over 240,000 shale wells.

Table 1: Descriptive Statistics: Bird Population and Species Counts

| Descriptive Statistics: Bird Population and Species Counts | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Bird Count (Winsorized) | | | Bird Count (Non-Winsorized) | | | Species Count (Non-Winsorized) | | |
| | Mean | Std. Dev. | Max | Mean | Std. Dev. | Max | Mean | Std. Dev. | Max |
| **Grassland/Shrubland** | 718.5 | 978.5 | 15,457 | 741.2 | 1,351.1 | 82,319 | 10.3 | 6.3 | 48 |
| **Woodland** | 1,783.8 | 1,860.1 | 24,187 | 2,251.9 | 30,088.8 | 5,008,455 | 24.5 | 7.5 | 62 |
| **Wetland** | 4,101.8 | 6,561.0 | 73,359 | 7,782.5 | 27,117.6 | 1,212,650 | 22.3 | 16.9 | 97 |
| **Other Habitat** | 4,024.1 | 4,019.8 | 32,315 | 15,986.2 | 198,523.8 | 12,500,803 | 14.2 | 7.0 | 51 |
| **Urban** | 7,159.4 | 6,261.2 | 52,425 | 20,079.3 | 201,404.1 | 12,507,177 | 35.0 | 10.8 | 71 |
| **Non-Urban** | 3,737.6 | 6,222.0 | 73,779 | 6,904.5 | 27,627.8 | 1,504,963 | 37.2 | 24.0 | 163 |
| **Resident** | 1,288.6 | 1,182.4 | 17,106 | 1,291.6 | 1,198.6 | 20,811 | 14.3 | 4.1 | 40 |
| **Short/Irruptive Migrants** | 8,075.1 | 8,391.9 | 79,298 | 21,844.8 | 202,490.9 | 12,507,816 | 47.6 | 22.4 | 132 |
| **Moderate/Long Migrants** | 1,575.8 | 3,035.8 | 38,916 | 3,954.8 | 22,706.7 | 1,501,440 | 10.5 | 9.9 | 73 |
| **Total** | 10,891.8 | 11,201.7 | 117,298 | 26,971.4 | 207,567.7 | 13,077,311 | 72.1 | 33.2 | 228 |

- Note: we removed min from this table for width purposes, but the mins were consistently at or near 0 as winsorizing at 99% did not affect the left tail.

Across the 2,234 circles nationally, Katovich constructed separate panels at the circle-year level for both shale and wind, each with 33,746 total records. The panels contain identifying geographic data, the number of wells or turbines built both annually and cumulatively, bird data by count and by species, and key covariates including weather conditions and land use data.

The outcomes of interest are the winsorized counts of birds and the non-winsorized counts of species by circle. Included in the bird data are the biome along with the migratory patterns. By winsorizing the counts at 99%, the extreme outliers are removed, bringing down the mean, standard deviation, and maximum values across the board. This accounts for some of the arbitrarily large round number counts as well as true outliers. Katovich conducted a robustness check against the raw data and found results to hold regardless. Table 1 captures the descriptive statistics for the key outcomes by category.

We recreated Figures 2 and 3, the maps of avian populations and energy infrastructure capacity building from the original paper. We were not able to recreate the figures exactly due primarily to dropped data from much of the Midwest and Southeast but found the coastal results matched Katovich's original figures. We think that this data was dropped due to county lines not aligning with CBC circles.

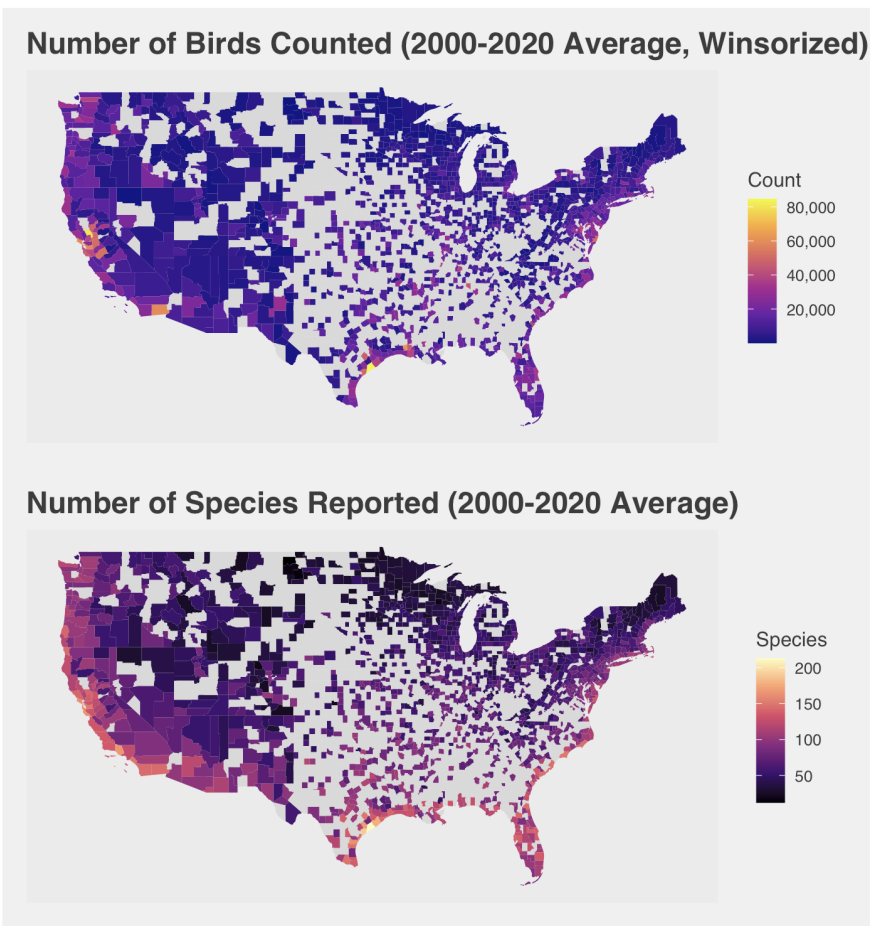



Figure 2 Replication

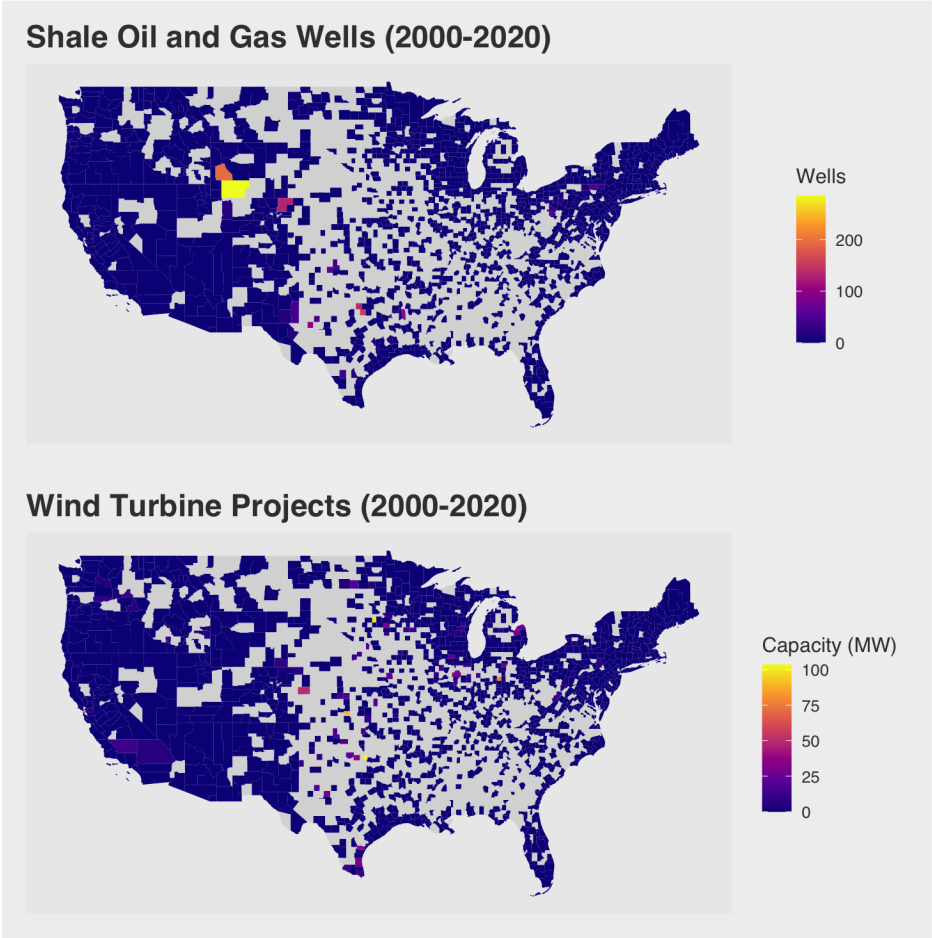



Figure 3 Replication

# 2 Analysis

## Parallel Trends

Katovich used separate difference-in-differences (DID) specifications on the first year of built energy infrastructure, either shale wells or wind turbines, to estimate the effects on the avian ecosystem.

The key identifying assumption for DID analyses is the parallel trends assumption (PTA). The PTA states that units in the treated group would have had the same change over time as the untreated group if the treatment group had counterfactually never been treated. Here, the PTA is that there would be no heterogeneous changes to the ecosystem if no energy infrastructure were ever built in the treated geographies. The ATT measures the change in trend (if any) from shale wells or wind turbines built within 5km of a CBC circle.

The primary threats to PTA here are driven by the heterogeneity in where energy infrastructure is built. The PTA requires that there are no other factors affecting the birds population and biodiversity, but some

human activities like city expansion can affect the birds. The effect of human activities and oil drilling are mixed, which could affect the causal identification of the ATT. Also, other weather situations could impact the bird populations. Therefore, we need to control for these factors in our specifications.

## Difference-in-Differences Analyses

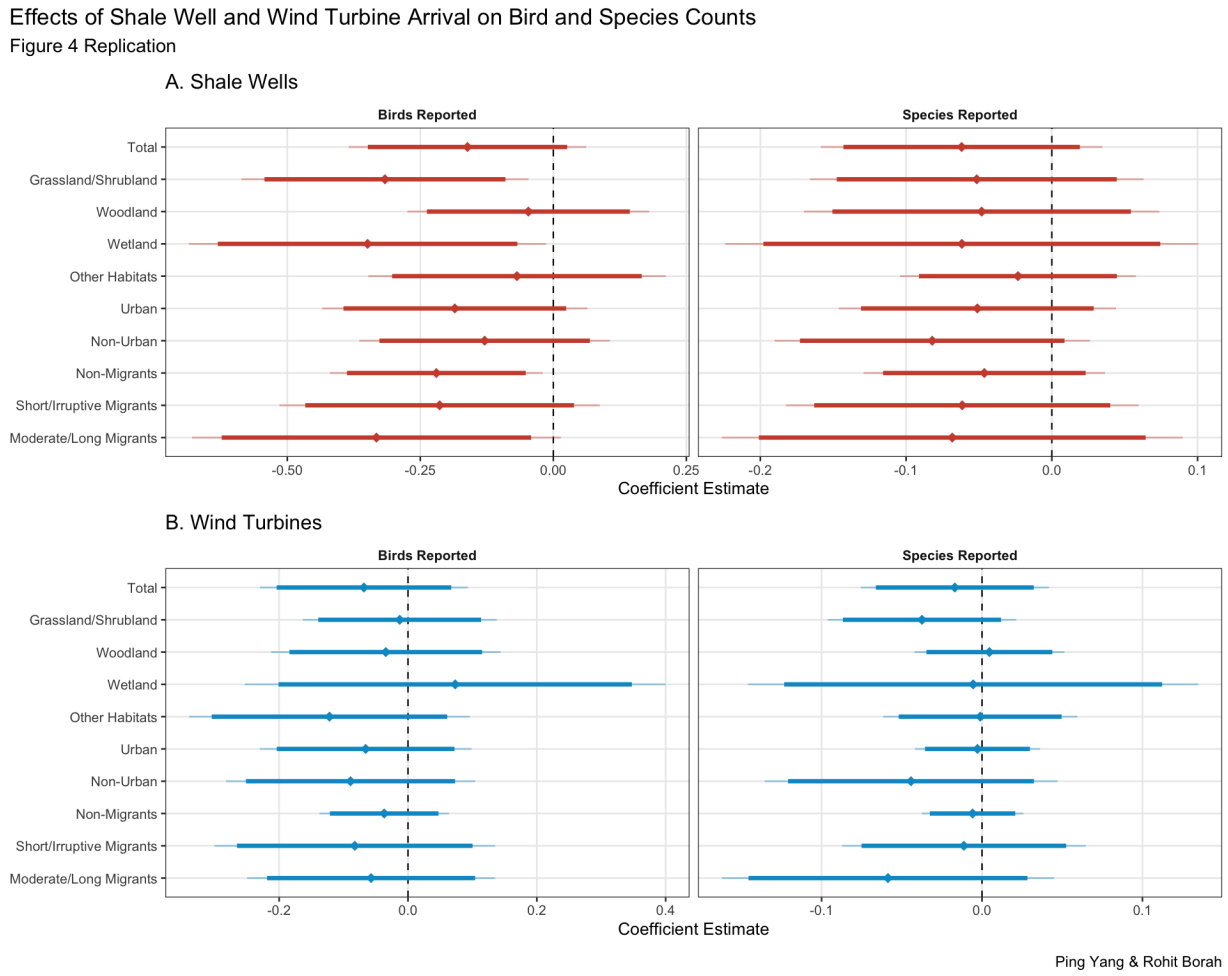
We replicated Figure 4 to visualize the point estimates and error bars around the different effect sizes by habitat and migratory types. The figure reports coefficient estimates with both 90% and 95% confidence intervals from difference-in-differences specifications that regress the number of birds or the species counted, total and disaggregated by bird characteristic, on relative period indicators both before and after the year shale wells were first drilled or wind turbines were first constructed within 5km of the border of a CBC circle. The control group consists of never-treated circles. Specifications include year and circle fixed effects and a vector of covariates including the number of counters, minimum/maximum temperature, snowfall, and wind speed in the circle on the day of the count, and proportions of the CBC circle occupied by agriculture, pasture, and developed land uses.

Standard errors are clustered at the circle level, and the model is estimated using Callaway and Sant'Anna's group-time difference-in-differences estimator. Continuous outcomes are transformed using the inverse hyperbolic sine function. We estimate both results with inverse probability weighting (IPW) estimation, aggregating results by calendar. To address potential selection bias and avoid linear extrapolation risks, we use IPW instead of the first-step regression estimator. While doubly robust was also considered, limited observations in year and cohorts make it numerically unstable.

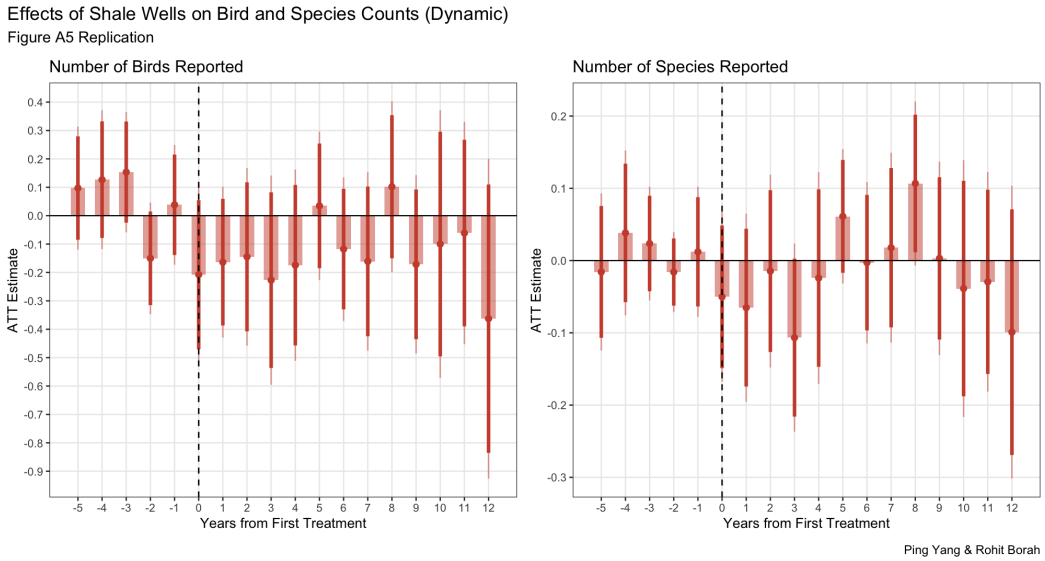
After conducting a difference-in-differences analysis in R, the results we found are mostly consistent with the conclusions found in the original paper. From Panel A, we observed that shale activity has a negative effect on both the total count of birds and the number of species reported. This is particularly noticeable in grassland, wetland, non-migrant, and longer migration populations. The total effect of shale wells on bird population count is $-16.1\%$, consistent with Katovich's original findings. As shown in Panel B, we do not find evidence that the construction of wind turbines negatively effects bird populations or biodiversity.

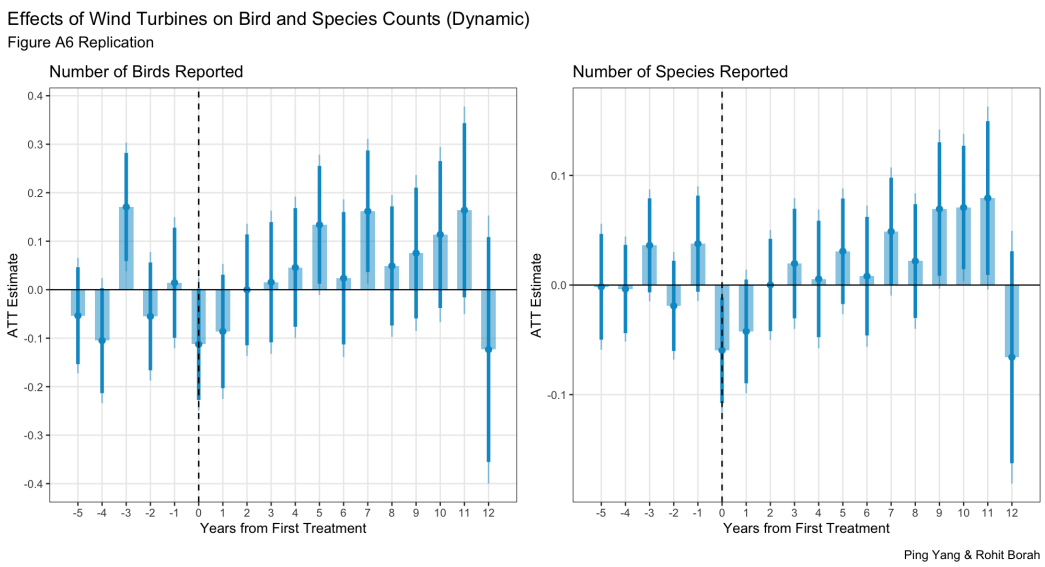
## Event Study Specification

For both shale and wind, pre-treatment effects cluster around 0 with wide error bars and no clear pattern prior to treatment year 0. We fail to reject the parallel pre-trends assumption based on the event study analysis. Shale does appear to more negatively impact the ecosystem than wind, which is consistent with our findings in the DID analysis.




This result is more specific to the number of birds reported where as the species count is more variable. These results are also consistent with Katovich's event study plots reported in the appendix of the original paper (Figures A5 and A6).




# 3 Extension

## DID Specification

We conducted the county-level analyses as an extension of the original paper. The county-level difference-in-differences specification is:

$$\widetilde{y}ct = \beta_1 Dct + \mathbf{X}ct'\beta_2 + \gamma_c + \delta_t + \varepsilon jt$$

- $\widetilde{y}_{ct}$ is the IHS-transformed county-year average of either bird or species count.
- $D_{ct}$ is the treatment indicator variable that is set to 1 whenever the first shale well or turbine is built in that county

- $\mathbf{X}_{ct}$ is the vector of counter, weather, and land use covariates at the county-year level
- $\gamma_c$ is the county fixed effect variable for all time-invariant county characteristics
- $\delta_t$ is the year fixed effect variable

Standard errors are clustered at the county level. The model is estimated using the Callaway-Sant'Anna DID estimator, which uses aggregate group-time average treatment effects for staggered treatment timing. The ATT for cohort $g$ at time $t$ using IPW is:

$$ATT(g,t) = \mathbb{E}\left[\left(\frac{\mathbf{1}(G_c = g)}{P(G_c = g)} - \frac{\frac{p(X_c)(1-G_c)}{1-p(X_c)}}{\mathbb{E}\left[\frac{p(X_c)(1-G_c)}{1-p(X_c)}\right]}\right)(Y_{ct} - Y_{c,g-1})\right]$$

- $p(X_c)$: The Propensity Score. This is the probability that a county is treated based on pre-treatment covariates.
- Weights: The control group (never-treated counties) is reweighted to match the characteristics of the treated cohort $g$, ensuring that the comparison is based on counties with similar "propensity" for shale and wind activity.
- $Y_{ct} - Y_{c,g-1}$: The change in the county-level average bird counts and species from the pre-treatment periods.

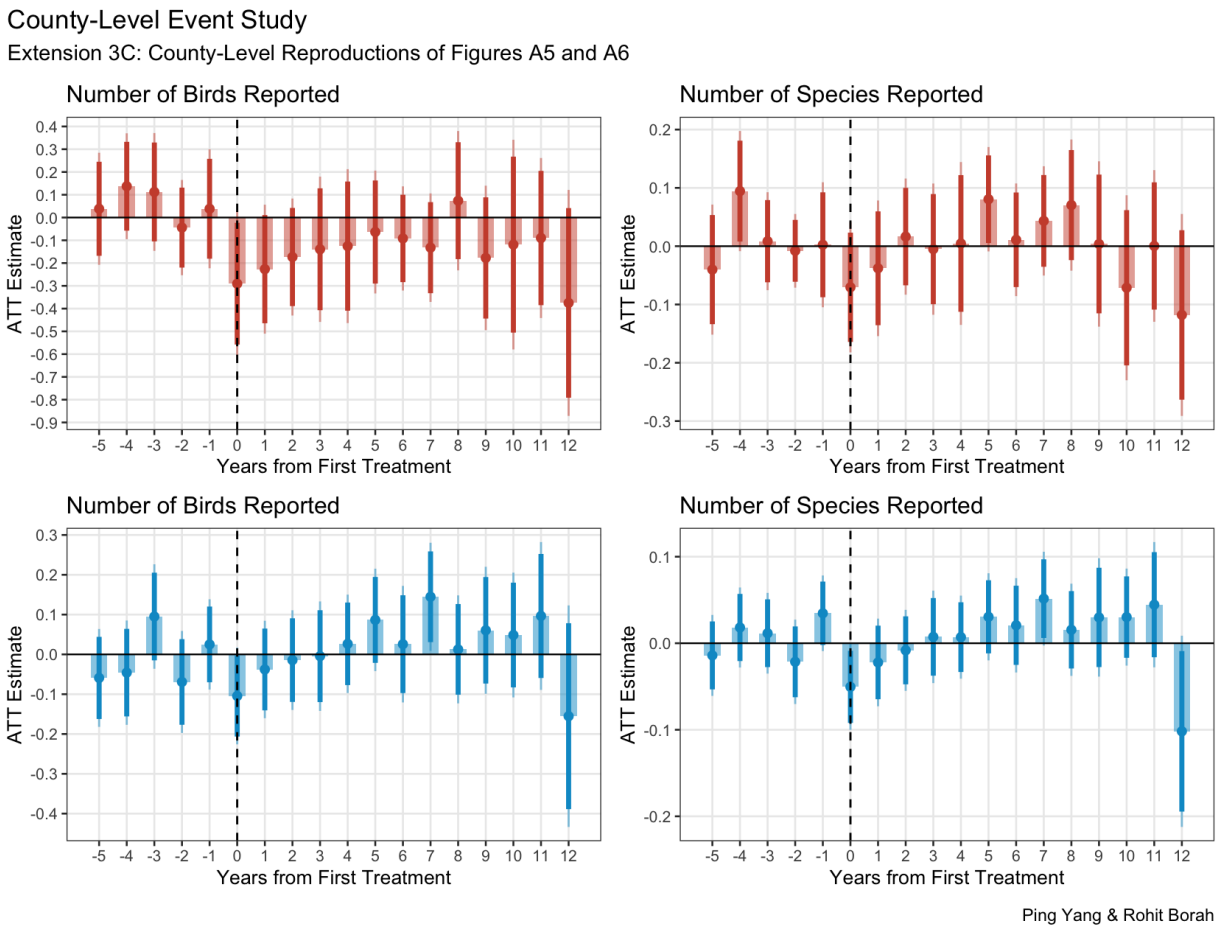
We again used IPW consistent with the methodology used in the circle-level analysis in Section 2. Causal identification relies on the average treatment effect on treated counties compared to the counterfactual outcome if a shale well and/or wind turbine had instead never been built with staggered treatment timing. Using the CS DID estimator allows us to better understand the variation in the effects of treatment timing on bird populations by avoiding the negative-weight bias common to two-way fixed effects estimators.

## Parallel Trends Assumption

The PTA states that the average bird population and species that received shale or wind activities would have followed the same trend as those did not conditional on observed covariates. Aggregating the data to the county level smooths out some noise found at circle level. By using IPW, we further strengthen the PTA by making the control group reweighted to match the treated groups' pre-treatment characteristics. We think that the PTA is more plausible at the county level because it absorbs some of the potential variance by often accounting for multiple CBC circles within one geography.

## County-Level Event Study

Outcomes are aggregated to the county-year level using the mean across CBC circles. This ensures that the estimated treatment effect reflects changes in average population density and richness, rather than fluctuations in the number of active sampling sites within a county. We conduct county-level Staggered DID using IPW, consistent with the methodology used in the circle-level analysis.




The effects of wind activities are consistent with the conclusions at the circle level previously conducted, but the shale outcomes have some interesting variation.

From the event study of shale construction on bird populations, we can see a sharp, immediate drop at year 0 followed by negative estimates for most time periods in next 12 years. The consistently negative dots suggest that shale activity is putting a steady downward pressure on bird population, which is consistent with the conclusions from the circle-level analysis. However, we do not find significant negative effect of shale activities on bird species. The point estimates bounce around the zero line throughout the entire post-treatment period, suggesting that the variety of species in a county is not systematically altered by the introduction of shale wells when measured at this aggregate scale.

We did find it interesting that the shale effects on the count of species reported disappeared when aggregated at the county level. Species richness is highly sensitive to the scale of observation. This insignificance suggests that while shale may drive a species out of a specific circle, it rarely drives a species out of an entire county. This makes the impact effectively invisible at the aggregate level.

# 4 Discussion

Our replication of the Katovich study roughly aligns with the original findings. We used the Callaway-Sant'Anna DID estimator with inverse probability weighting to determine the causal impact of shale wells and wind turbines on both bird populations and biodiversity. We found that digging shale wells has a directionally negative effect on bird populations, although not to the same level of significance or precision as Katovich did originally. We also found that building wind turbines has no measurable effect on bird populations or species counts. This aligned cleanly with the original analysis. Using event-study, we found that the parallel trends assumption held for this data.

As an extension of the original analysis, we also conducted a DID on the impact of shale wells and wind turbines at the county level and found that the outcomes were consistent with . Overall, our replication confirms the conclusions from the original paper that digging shale oil and gas wells harms birds while wind turbines do not.